

Name: _____

Date: _____

Class: _____

MyDesign Portfolio

Element C Initial Template

Element C: Presentation and Justification of Solution Design Requirements

Design Requirement #1 (HIGH Priority)

Our first design requirement is to make a successful detection system that detects fire and can detect the fire in an efficient amount of time. We can make something similar to a smoke detector by sensing smoke particles in the air.

Design Requirement #2 (HIGH Priority)

Create a successful suppression system that works like a sprinkler system by receiving the signal from the detection system and letting water out to suppress the fire and stop the fire from potentially spreading and burning the whole apartment down.

Design Requirement #3 (HIGH Priority)

Create a notification system that notifies that a fire is happening with a loud alarm that receives the signal from the detection system so it can make that alarm and that it is making the alarm only while a fire is present.

Design Requirement #4 (HIGH Priority)

Create a house that can stand on its own so that the apartment does not tip over and possibly damage the suppression or detection system which would result in a failed product.

Design Requirement #5 (HIGH Priority)

Create furniture inside the apartment to make the apartment livable by adding furniture inside the apartment that is not made of non-flammable materials so we can see if the suppression and detection system really works and if an actual person would live in that apartment they would actually want to live in that apartment.

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Design Requirement #6 (LOW Priority)

Make the apartment visually appealing so that people may be drawn to our design and see that it may become interesting.

Design Requirement #7 (LOW Priority)

Find an optimal place to put our systems so they are safe and they have the ability to suppress the fire and detect the fire. Also so none of our systems are damaged while the fire is present.

Rubric

Element C: Presentation and Justification of Solution Design Requirements						
C1. Determination of design requirements. _____ design requirements are related to prior research.	score = 5 80%-100% of	score = 4 60%-79% of	score = 3 40%-59% of	score = 2 20%-39% of	score = 1 1%-19% of	score = 0 No
C2. Requirements list. Design requirements _____.	score = 5 are listed, prioritized, clear, and detailed	score = 4 are listed, prioritized, generally clear, and detailed	score = 3 are listed, prioritized, generally clear, and somewhat detailed	score = 2 are listed and prioritized but some or all may be incomplete and/or lack specificity	score = 1 are listed, formatted, and prioritization is attempted, but these may be partial or overly general	score = 0 are not present or are too vague
C3. Quantity of high-priority design requirements. _____ of the design requirements listed are identified as high priority.	score = 5 Five	score = 4 Four	score = 3 Three	score = 2 Two	score = 1 One	score = 0 None

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	score = 5	score = 4	score = 3	score = 2	score = 1	score = 0
C4. Objective & measurable. Design requirements _____.	are consistently objective and measurable	are nearly always objective and measurable	are generally objective and measurable	may be only sometimes objective and/or measurable	are insufficiently measurable	are not measurable
	score = 5	score = 4	score = 3	score = 2	score = 1	score = 0
C5. Leading to solution. Design requirements _____.	would be highly likely to lead to a tangible and viable problem solution	would be likely to lead to a tangible and viable problem solution	have a potential to lead to a tangible and viable problem solution	have some potential to lead to a tangible and viable problem solution	have little potential to lead to a tangible and viable problem solution	cannot lead to a tangible and viable problem solution
	score = 5	score = 4	score = 3	score = 2	score = 1	score = 0
C6. Primary stakeholder validation. There is evidence that requirements represent the needs of, and have been validated by, _____ primary stakeholder groups.	80%-100% of	60%-79% of	40%-59% of	20%-39% of	1%-19% of	no primary stakeholder groups.